

Thomas VAUDESCAL

Quantitative Developer | Quantitative Researcher

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Profile

Quantitative researcher and developer who takes options research from hypothesis to production code. MSc thesis (A+) on dealer gamma exposure and intraday volatility, now a **working paper in preparation for the International Review of Financial Analysis (IRFA)**: it replicates and extends Dim, Eraker and Vilkov (2025) on public data over 13,411 SPX bars. Builds the tooling behind it in **Python, Numba and JAX**: a backtesting platform that runs five years of SPX intraday option chains (216M rows) in under a minute, and a pricing platform with Greeks by adjoint algorithmic differentiation. Research discipline is enforced in code: pre-registered hypotheses, a 66-trial registry, deflated Sharpe ratio and probability of backtest overfitting. MSc in Financial Engineering, HEC Montréal (**GPA 3.88/4.3**).

Education

HEC Montréal | MSc in Financial Engineering | GPA 3.88/4.3

Montréal | August 2022 - December 2024

- **Thesis, A+ with commendation from the examination committee:** Gamma Exposure and Intraday Volatility Dynamics in the SPX Options Market, later applied to a systematic options strategy.
- **Scholarships of excellence:** Deloitte M.Sc., Alpha Fixe Capital, Jocelyne & Jean C. Monty I and II. **Competitions:** National Bank challenge winning team at ConUHacks VIII (2024), top 10 of 66 teams at the McGill FIAM Portfolio Management Hackathon, 2nd place at the HEC Montréal Trading Competition.

HEC Montréal | BBA in Mathematics, Finance & Economics

Montréal | August 2017 - May 2022

- **GPA:** 3.8/4.3 in specialization courses. Minor in Mathematics at Université de Montréal (probability, statistics, stochastic processes, real analysis, C/C++).

Research & Projects

Gamma Exposure and Intraday Volatility | MSc thesis and IRFA working paper

Private repository

- Replicates and extends Dim, Eraker and Vilkov (2025) with public ORATS data: 13,411 thirty-minute SPX bars (2020 to 2025), cross-checked on SPY and QQQ, with walk-forward validation. The non-ODTE gamma coefficient is -0.442 on public data, against -0.229 in the original dealer-data study.
- 83-page thesis and a 53k-line Python pipeline (79 tests). No number in the manuscript is hard-coded, each table comes from machine-readable outputs.

OptiBack | Options backtesting and research platform

Private repository

- Numba struct-of-arrays engine on ORATS intraday option chains. Backtests five years of SPX (216M rows, 96.5k snapshots) end to end in under a minute on one workstation. 49k lines of source, 2,448 automated tests.
- Research protocol enforced in code: pre-registered hypotheses, a 66-trial registry, deflated Sharpe ratio on an honest and effective number of trials, probability of backtest overfitting (CSCV), walk-forward windows and synthetic stress scenarios.

Derivatives pricing platform | JAX and Numba

Private repository

- Analytic, FFT and Monte Carlo pricers, Heston, Bates, Merton and GARCH calibration, exotic and structured-product catalog. Greeks by adjoint algorithmic differentiation on JAX cost about 3x one price (GBM, 500k paths), run 2.9x faster than bump-and-reprice and stay within 0.25% of Black-Scholes. 69k lines of source, 2,329 automated tests.

Options-flow capture | Python, DuckDB, systemd

Private repository

- Market-hours poller for SPX and SPY gamma, flow and dark-pool data, stored in a 5 GB DuckDB database and supervised by systemd with a watchdog. 320 automated tests.

Ganesha | ganesha.synergo.com

Synergo | Production

- Multi-client retrieval-augmented assistant in production (Python, FastAPI, Qdrant, Next.js). On a 100-question controlled evaluation, 109 of 110 claims were supported by their cited sources, with a mean latency of 7.7 s.

Public code | github.com/Thomas4390

Open source

- quant-paper-lab (finance papers reproduced on public data) and fx_strategies (walk-forward framework).

Professional Experience

Synergo | Founder & Quantitative Researcher

Montréal | May 2025 - Present

- Designed, backtested and **deployed a systematic SPX options strategy to live trading**, turning MSc research on dealer gamma exposure into a production signal and execution workflow.
- Validated FX and gold strategies for a client under a frozen protocol (purged walk-forward, bootstrap, holdout read once). **Rejected 45 variants and recommended against deployment.**
- Developed **Rust execution components** for market-data ingestion, quote updates and order routing, and profiled their latency against the Python research stack.
- Prototyped systematic options market making with skew-aware quoting, inventory-based price adjustments and microstructure-based adverse-selection filters.

Deloitte | Quantitative Analyst, Financial Risk

Montréal | October 2024 - June 2025

- Developed and validated **credit risk models (PD, LGD, EAD)** for major Canadian financial institutions under IFRS 9 and Basel III/IV, leading independent validation through statistical backtesting, sensitivity analysis and stress testing.
- Built **automated monitoring dashboards and validation pipelines in Python**, reducing manual review time by **40%**.
- Authored validation reports for **senior stakeholders and client risk committees** (OSFI context).

National Bank Financial Markets | Quantitative Developer Intern

Montréal | May 2023 - December 2023

- In Global Equity Derivatives R&D, developed and deployed low-volatility and momentum **SmartBeta strategies covering \$800M in AUM** within a \$7B structured products division, with constrained portfolio optimization (cvxpy).
- Built an **end-to-end Python backtesting framework**: signal generation, portfolio construction, transaction costs, risk analytics and performance attribution.
- Delivered **\$1M in annual cost savings** through strategy optimization and migration off an external vendor.
- Worked with **structuring and trading teams**, and delivered performance and risk reports to the desk and portfolio managers.

Videns Analytics | Data Analyst

Montréal | April 2020 - June 2021

- Led **consulting engagements in insurance analytics** (predictive modeling, automated reporting).
- Designed a **Python and machine learning curriculum** adopted by UQAC. Promoted from intern to full-time.

Academic Involvement

Trading Club | Vice President Research

August 2023 - May 2024

- Led a team of 4 reproducing high-frequency FX and electricity-price papers, backtested in Python.

BNI-HEC Investment Fund | Quantitative Analyst

September 2022 - May 2023

- R&D team of a \$5M student-managed portfolio: quantitative models, dashboards, attribution and risk analysis.

Data Science Committee | Vice President Education

August 2020 - August 2021

- Organized the "AI in 2020" conference (40+ attendees), Python workshops and machine learning seminars.

Teaching Assistant | Intermediate Microeconomics, Mathematics Preparatory Year

August 2021 - May 2023

- Corrected exams, held consultation hours and created exercises (200+ students).

Technical Skills & Interests

Programming	Python, C++, Rust, SQL, R, MATLAB.
Computer Science	Data structures and algorithms, object-oriented design, event-driven systems, concurrency (asyncio), latency profiling, performance optimization.
Python Stack	NumPy, Numba, JAX, pandas, Polars, DuckDB, SciPy, scikit-learn, LightGBM, PyTorch, cvxpy, QuantLib.
Quant Finance	Derivatives pricing, Greeks via AAD, Monte Carlo, calibration (Heston, Bates, Merton, GARCH), backtesting, walk-forward validation.
Risk Modeling	Credit risk (PD, LGD, EAD), model validation, stress testing, model monitoring.
Data & Tools	Bloomberg Terminal & API, AWS, Docker, Git, Linux, systemd, FastAPI, PostgreSQL, TypeScript, REST and WebSocket APIs.
Languages	French (native), English (professional proficiency).
Interests	Chess (2000 Elo, Chess.com rapid) , mathematics, philosophy, programming. Marathon running, track and field, mountain hiking, indoor climbing, swimming, scuba diving.